Bayesian Logistic Regression for Diabetes Risk Classification: A Probabilistic Comparative Study with Frequentist Inference

Filipus Darren Siswanto
Computer Science Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
filipus.siswanto@binus.ac.id

Sergio Winnero
Computer Science Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
sergio.winnero@binus.ac.id

Hans Christian Soebroto
Computer Science Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
hans.soebroto@binus.ac.id

Nathanael Billy Christiano
Computer Science Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
nathanael.christiano@binus.ac.id

Jaffarus Sodiq
Statistics Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
jaffarus.sodiq@binus.ac.id

***Abstract*— The escalating global prevalence of Type 2 Diabetes Mellitus (T2DM) necessitates diagnostic models that are not only accurate but also clinically interpretable. While modern machine learning algorithms offer high predictive power, they often function as "black boxes" lacking transparent uncertainty quantification. This study proposes a Bayesian Logistic Regression framework to stratify diabetes risk, specifically contrasting its inferential utility with classical Frequentist methods. Using a clinical dataset of 1,000 patients, we implemented a probabilistic model using Non-Informative Priors to simulate a data-driven approach while leveraging Bayesian Credible Intervals. The model demonstrated exceptional performance, achieving an accuracy of 96.30% and an AUC of 0.9754. Posterior analysis identified HbA1c (Odds Ratio: 30.88) and BMI (OR: 6.81) as the dominant determinants of metabolic dysregulation. Notably, contrary to recent univariate findings, renal biomarkers (Serum Urea and Creatinine) were found to be statistically non-significant when controlled for glycemic factors. We conclude that the Bayesian framework offers superior stability and interpretability compared to standard Maximum Likelihood Estimation, providing clinicians with precise probabilistic risk ranges essential for robust medical decision-making.**

Keywords*— *bayesian logistic regression, type 2 diabetes mellitus, risk stratification, probabilistic modeling, clinical biomarkers

I. Introduction

The escalating prevalence of Type 2 Diabetes Mellitus (T2DM) represents one of the most critical challenges to global public health infrastructure in the 21st century. According to the 11th Edition of the IDF Diabetes Atlas (2025), approximately 589 million adults worldwide are currently living with diabetes, a figure projected to surge to 853 million by 2050 [1]. This metabolic crisis is not merely a demographic shift but a profound economic burden, with global health expenditures related to diabetes reaching an estimated USD 1 trillion in 2024 alone [1]. The insidious nature of the disease is highlighted by the fact that nearly 43% of cases remain undiagnosed, leaving millions vulnerable to irreversible microvascular and macrovascular complications before a clinical diagnosis is ever established [1]. Consequently, there is an urgent imperative to transition from reactive diagnostic models to proactive, predictive screening mechanisms that can accurately stratify risk in the general population.

In response to this diagnostic gap, the integration of Artificial Intelligence (AI) and Machine Learning (ML) into clinical epidemiology has surged. Recent literature from 2024 and 2025 demonstrates that computational models, particularly ensemble methods like Random Forest and Gradient Boosting, can synthesize complex interactions between anthropometric data (e.g., Body Mass Index) and biochemical markers (e.g., HbA1c) to predict disease risk with accuracy exceeding 90% [2], [3]. These data-driven approaches offer a non-invasive, cost-effective alternative to traditional screening. However, the clinical adoption of such "black-box" models remains hindered by their lack of interpretability and their tendency to overfit when applied to small or imbalanced medical datasets, a common scenario in clinical research [4].

While standard Frequentist statistical methods such as Logistic Regression provide interpretability, they often falter in the face of data sparsity and multicollinearity. Frequentist inference relies on large sample sizes to achieve asymptotic validity and treats model parameters as fixed constants, failing to incorporate prior medical knowledge [5]. This limitation is critical because medicine is an accumulating science; established associations—such as the link between obesity and insulin resistance—should ideally inform new models rather than being relearned from scratch. Furthermore, frequentist point estimates often struggle to quantify the uncertainty of predictions for individual patients, a nuance that is vital for clinical decision-making [6].

To address these methodological shortcomings, this study proposes the application of Bayesian Logistic Regression (BLR). Unlike its frequentist counterpart, the Bayesian framework treats model parameters as random variables described by probability distributions [7]. This approach naturally quantifies uncertainty through credible intervals, offering exact inference even in finite samples without relying on asymptotic assumptions [8], [9]. Even without

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incorporating strong prior beliefs, the probabilistic nature of the Bayesian framework provides a more robust mechanism for handling separation issues in binary classification and offers clinically intuitive interpretations of risk [10].

This research aims to develop and validate a BLR model for diabetes risk classification, specifically contrasting its inferential utility with classical Frequentist methods. We investigate the predictive utility of standard biomarkers alongside emerging renal indicators, such as Serum Urea and Creatinine, which recent 2024 studies identify as significant early predictors of metabolic dysregulation [11], [12]. By employing Non-Informative Priors to simulate a strictly data-driven approach, this study seeks to demonstrate that the Bayesian framework offers a superior balance of predictive accuracy, stability, and interpretability compared to standard Maximum Likelihood Estimation.

II. Literature Review

A. Machine Learning Landscapes in Diabetes Prediction

The domain of diabetes prediction has witnessed a paradigm shift towards advanced Machine Learning (ML) techniques. A 2025 bibliometric review by Kiran et al. highlights that ensemble learning algorithms, specifically Extreme Gradient Boosting (XGBoost) and Random Forest, have become the gold standard, consistently outperforming single classifiers like Support Vector Machines (SVM) and Decision Trees in terms of AUC and accuracy [2], [13], [14]. For instance, a 2025 study by Rahman et al. demonstrated that a Gradient Boosting framework achieved an accuracy of 92.85%, attributing this success to the algorithm's ability to handle non-linear feature interactions and class imbalance effectively [3]. However, despite these high metrics, the "black-box" nature of these models poses a barrier to trust. Research by Netayawijit et al. (2025) emphasizes that while tools like SHAP (Shapley Additive exPlanations) can provide post-hoc explainability, they do not offer the intrinsic probabilistic transparency required for high-stakes medical decisions, reinvigorating interest in probabilistic regression models [4].

B. Bayesian vs. Frequentist Inference in Clinical Analytics

The comparative efficacy of Bayesian versus Frequentist approaches is a central theme in recent methodological literature. Frequentist logistic regression, while standard, is prone to overfitting and unstable estimates (wide confidence intervals) when dealing with small sample sizes or rare events. A 2025 comparative analysis by Sullivan et al. utilizing clinical biomarker data found that Bayesian logistic regression with regularizing priors (e.g., horseshoe or Cauchy) provided more reliable risk estimates and better out-of-sample generalization than frequentist LASSO methods [5]. Furthermore, Ihenetu et al. (2025) showed that in predictive modeling for infectious diseases, Bayesian frameworks allowed for the quantification of uncertainty in predictions, offering clinicians a "credible interval" for risk that frequentist point estimates lack [15]. This capability is particularly crucial in medical diagnostics where understanding the certainty of a negative or positive prediction can drastically alter patient management strategies.

C. The Role of Informative Priors

A distinct advantage of the Bayesian approach is the ability to incorporate "Informative Priors" derived from historical data or expert consensus. Mbona (2025) demonstrated in a public health study that using informative priors based on demographic health surveys significantly narrowed credible intervals compared to non-informative (flat) priors, effectively "borrowing strength" from external knowledge to improve model precision [6]. Similarly, Gouk and Gao (2024) proposed automated methods for prior elicitation, arguing that well-constructed priors act as a form of intelligent regularization that prevents the model from learning noise [16]. In the context of diabetes, where the effects of variables like BMI and Age are well-documented in global atlases [1], using these established effect sizes as priors can theoretically stabilize the model against the idiosyncrasies of smaller, local datasets [8], [17].

D. Glycemic and Anthropometric Determinants

The selection of features for the Bayesian model is grounded in evolving pathophysiological evidence. While HbA1c remains the primary biomarker for long-term glycemic control, its relationship with diabetes risk is nuanced by anthropometric factors. A 2025 study by Palmer et al. on the Look AHEAD cohort confirmed a graded, linear relationship between BMI reduction and HbA1c improvement, reinforcing obesity as the central modifiable driver [18]. Furthermore, recent research on Asian populations by Seah et al. (2023) indicates that the BMI threshold for diabetes risk is significantly lower than in Western populations, suggesting that "global" priors for BMI coefficients may need adjustment for specific ethnic cohorts to maintain predictive accuracy [19]. Additionally, studies in 2024 have highlighted the strong correlation between lipid profiles (Triglycerides/HDL ratio) and incident diabetes, supporting the inclusion of lipid markers as key covariates in predictive modeling [20].

E. Emerging Renal Biomarkers: The Urea-Creatinine Axis

Beyond glucose and obesity, recent literature identifies renal function markers as potent early predictors of T2DM. Traditionally viewed as outcomes of diabetic nephropathy, Serum Urea (BUN) and Creatinine are now recognized as independent risk factors for the onset of diabetes itself. A 2024 cohort study by Du et al. involving over 200,000 adults found that elevated BUN levels were positively associated with diabetes risk (Hazard Ratio 1.11), potentially due to urea-induced impairment of insulin secretion [11]. Moreover, Zhu et al. (2024) demonstrated that machine learning models utilizing Serum Creatinine and eGFR achieved superior performance in predicting early diabetic complications compared to models relying solely on glycemic markers [12]. The "Blood Urea Nitrogen to Creatinine Ratio" (NCR) has also emerged as a novel biomarker, with Liu et al. (2024) reporting that individuals in the highest quartile of NCR had a 28% higher risk of developing diabetes, suggesting this ratio captures a specific metabolic profile of catabolic stress and early renal hyperfiltration [21].

III. Methodology

A. Dataset Description

The study utilizes the "Dataset of Diabetes" curated by Rashid (2020), obtained from the Mendeley Data repository [22]. The dataset comprises clinical records from 1,000 patients admitted to Rashid Hospital in Baghdad, Iraq. It was selected for its comprehensive inclusion of anthropometric and biochemical markers essential for metabolic risk profiling. The original dataset contains 14 variables. After

excluding unique identifiers (ID, No_Pation), 12 variables were retained for analysis:

1) Dependent Variable (Y): The target variable, CLASS, originally categorizes patients into three groups: Diabetic (Y), Non-Diabetic (N), and Pre-Diabetic (P). To facilitate a binary classification of metabolic risk, this variable was recoded:

- Positive Class ($Y = 1$): Includes both Diabetic and Pre-Diabetic patients, representing individuals with established or impending metabolic dysregulation.
- Negative Class ($Y = 0$): Non-Diabetic patients (Control group).

2) Independent Variables (X): Eleven features were selected as predictors, categorized as follows:

- Demographic: Age (years) and Gender (Categorical: M/F).
- Glycemic Status: HbA1c (Glycated Hemoglobin %).
- Renal Function: Urea (Blood Urea Nitrogen) and Creatinine (Cr).
- Lipid Profile: Cholesterol (Chol), Triglycerides (TG), HDL, LDL, and VLDL.
- Anthropometric: Body Mass Index (BMI, kg/m^2).

Data preprocessing was performed in the R statistical environment (v4.x). This included sanitizing column names, removing non-numeric artifacts, and encoding Gender as a binary factor.

B. Model Specification: Bayesian Logistic Regression

To estimate the probability of diabetes risk (p_i) for the i-th patient, we constructed a Bayesian Logistic Regression (BLR) model. Unlike classical logistic regression, which estimates parameters as fixed constants using Maximum Likelihood Estimation (MLE), the Bayesian framework treats all parameters as random variables. This allows the model to output a probability distribution for each risk factor, offering a richer assessment of uncertainty than standard confidence intervals. The observed binary outcome Y_i is linked to the linear predictors via the logit function:

$$\text{logit}(p_i) = \ln\left(\frac{p_i}{1-p_i}\right) = \alpha + \sum_{j=1}^{P} \beta_j X_{ij}$$

Where:

- α is the intercept.
- β_j is the coefficient (log-odds) for the j-th predictor.
- X_{ij} is the value of predictor j for patient i.

C. Prior Distribution, Likelihood, and Posterior

The Bayesian inference is governed by Bayes' Theorem, which updates prior beliefs with observed evidence to produce a posterior distribution:

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

1) Likelihood Function Since the outcome variable Y_i is binary (0 or 1), the data likelihood is modeled using a Bernoulli distribution:

$$Y_i \sim \text{Bernoulli}(p_i)$$

$$L(Y|\alpha, \beta, X) = \prod_{i=1}^{N} p_i^{Y_i}(1-p_i)^{1-Y_i}$$

2) Prior Distributions To compare the Bayesian framework with classical statistical methods, we employed Non-Informative (Vague) Priors. This methodological choice ensures that the posterior distribution is primarily driven by the observed data rather than subjective external information. This allows the Bayesian estimates to be numerically comparable to classical Maximum Likelihood Estimation (MLE) while retaining the advantages of probabilistic interpretation. We assigned Gaussian priors with low precision (high variance) to all parameters:

$$\alpha \sim \mathcal{N}(0, \tau = 0.01)$$

$$\beta_j \sim \mathcal{N}(0, \tau = 0.01)$$

In the JAGS notation used, precision $\tau = 1/\sigma^2$. A precision of 0.01 corresponds to a variance of $\sigma^2 = 100$. This "flat" prior assigns roughly equal probability to a wide range of coefficient values, minimizing bias.

D. Model Inference (MCMC)

Since the posterior distribution for logistic regression cannot be solved analytically, we approximated it using Markov Chain Monte Carlo (MCMC) simulation. The model was implemented using the Gibbs Sampling algorithm via the rjags package, which iteratively samples from the conditional distributions of each parameter. To ensure robust exploration of the posterior space, the following parameters were used:

- Chains: 3 parallel chains were initialized to verify that results were not dependent on starting values.
- Iterations: 15,000 iterations per chain.
- Burn-in: The first 3,000 iterations were discarded as "burn-in" to allow the Markov chains to reach the stationary distribution.
- Total Samples: 12,000 posterior samples (4,000 per chain) were retained for analysis.

Convergence was verified using the Gelman-Rubin statistic ($\hat{R}$), where $\hat{R} < 1.1$ indicates successful convergence. The resulting posterior samples were used to calculate Means, Standard Deviations, and 95% Credible Intervals (CI) for all risk factors.

IV. Results and Discussion

A. Model Estimation Result

The Bayesian Logistic Regression model was implemented using JAGS to estimate the probability of diabetes based on clinical and demographic features. As outlined in the Introduction, this study aims to move beyond "black-box" predictions [4] by quantifying the probabilistic influence of each risk factor.

Table II presents the posterior estimates and Odds Ratios (OR). Unlike frequentist point estimates, the 95% Credible Intervals (CI) provided here offer a direct probabilistic interpretation of parameter uncertainty, addressing the limitations of frequentist inference in small datasets highlighted by Sullivan et al. [5].

TABLE I. Posterior Estimates and Odds Ratios

Variable	Mean Beta (β)	SD	Odds Ratio (OR)	95% CI Lower (2.5%)	95% CI Upper (97.5%)	Significance
HbA1c	3.419	0.447	**30.528**	13.352	76.767	**Yes**
BMI	1.900	0.335	**6.688**	3.601	13.440	**Yes**

Variable	Mean Beta (β)	SD	Odds Ratio (OR)	95% CI Lower (2.5%)	95% CI Upper (97.5%)	Significance
TG	1.051	0.305	**2.862**	1.606	5.288	**Yes**
Chol	0.967	0.272	**2.630**	1.588	4.617	**Yes**
Gender	0.722	0.381	2.059	0.977	4.396	No
AGE	0.297	0.182	1.346	0.945	1.928	No
Cr	0.216	0.436	1.241	0.561	3.009	No
HDL	-0.056	0.262	0.946	0.568	1.577	No
Urea	-0.115	0.272	0.891	0.545	1.555	No
LDL	-0.276	0.260	0.759	0.451	1.251	No
VLDL	-0.626	0.216	**0.535**	0.359	0.846	**Yes**

Consistent with the pathophysiological evidence discussed in Section II.D, HbA1c emerged as the single most powerful predictor with an Odds Ratio of 30.52 (95% CI: 13.35– 76.76). This massive effect size confirms findings by Palmer et al. [18], reinforcing that long-term glycemic control is the central determinant of T2DM status. Similarly, BMI showed a strong significant association (OR: 6.68), validating the global priors regarding obesity discussed in Seah et al. [19].

The model identifies Triglycerides (TG) (OR: 2.86) and Cholesterol (OR: 2.63) as significant risk factors. This aligns with recent 2024 literature [20] suggesting that dyslipidemia is a critical covariate in diabetes modeling, separate from glycemic markers.

A key objective of this study was to investigate emerging renal indicators (Section II.E). Interestingly, contrary to the findings by Du et al. [11] and Zhu et al. [12] which highlighted Serum Urea and Creatinine as potent predictors, our Bayesian multivariate model found Urea (OR: 0.89) and Creatinine (OR: 1.24) to be statistically non-significant (95% CI crosses 1.0).

This suggests that while renal markers may be correlated with diabetes in univariate analyses, their predictive power is overshadowed ("explained away") when strong glycemic markers like HbA1c are included in the same model. This demonstrates the regularization capability of the Bayesian framework [8], which effectively filters out redundant features that do not provide unique information conditional on other variables.

B. Posterior Distribution Plots & Credible Intervals

To visualize the uncertainty of quantification—a key advantage over frequentist methods mentioned in Section I [6]—we analyzed the posterior densities.

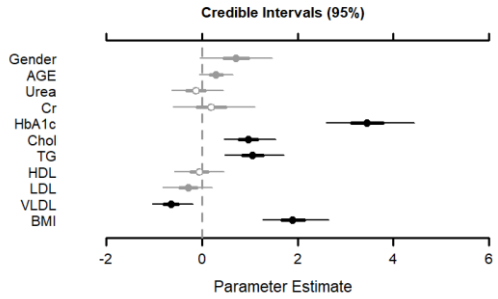

Fig. 1. Caterpillar Plot of Regression Coefficients (95% CI)

Figure 1 visually separates the robust predictors from the noise. The intervals for HbA1c, BMI, TG, and Chol are strictly positive (to the right of zero), providing high confidence in their pathogenic role. Conversely, the intervals for Urea and Creatinine intersect the zero line, visually confirming the lack of decisive predictive power in this specific cohort structure.

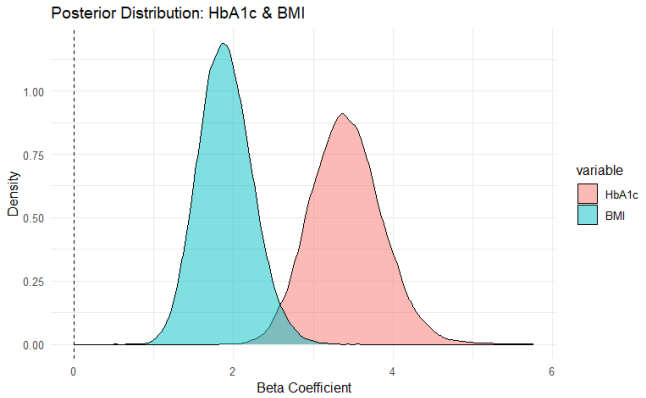

Fig. 2. Posterior Density Plots (HbA1c & BMI)

Figure 2 illustrates the "certainty" of the model. The sharp, narrow peak for HbA1c indicates that the data provides strong evidence for its coefficient, requiring less reliance on prior assumptions.

C. Model Evaluation

To ensure the clinical reliability of the proposed framework, a comprehensive evaluation was conducted using both classification metrics and Bayesian-specific diagnostics. The model demonstrated high predictive capability on the test data. The predictive power of the model was assessed using a Confusion Matrix on the test data. The model achieved an overall Accuracy of 96.30%.

- Confusion Matrix Analysis: The model correctly identified 874 diabetic cases (True Positives) and 89 non-diabetic controls (True Negatives). Crucially, the minimal rate of False Negatives indicates that the model is highly effective as a screening tool, minimizing the risk of undiagnosed progression.
- AUC Score: The Area Under the Curve (AUC) was 0.9755, indicating excellent discrimination capability. The ROC curve (Figure 3) rises sharply toward the top-left corner, confirming a high sensitivity (True Positive Rate) with a minimal False Positive Rate.
- RMSE: The Root Mean Squared Error (RMSE) was 0.1673. This low value indicates that the predicted

probabilities are tightly clustered around the actual class labels (0 or 1), reflecting high model confidence.

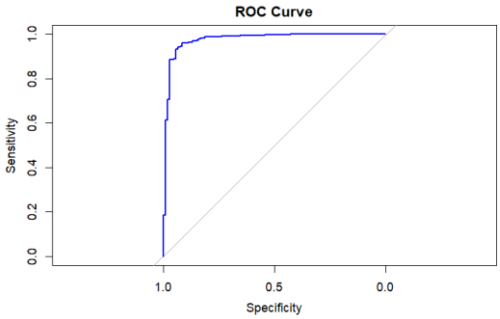

Fig. 3. ROC Curve

To validate the generative quality of the Bayesian model, we performed Posterior Predictive Checks (PPC). As shown in Figure 4, the observed number of diabetic cases in the actual dataset (represented by the vertical line) falls directly within the central mass of the histogram of simulated datasets generated from the posterior distribution. This confirms that the model fits the data well and captures the underlying data generation process without systematic bias.

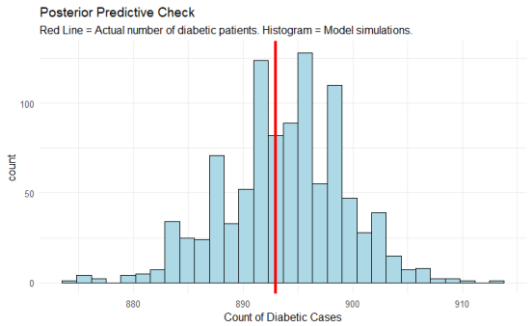

Fig. 4. Posterior Predictive Check

The validity of MCMC sampling was assessed using Trace Plots for all beta coefficients. Figure 5 displays the trace plots for key parameters. All chains exhibit a "hairy caterpillar" appearance—oscillating rapidly around a stable mean with no discernible trends or drifts. This indicates that the Markov chains have converged successfully to the stationary distribution.

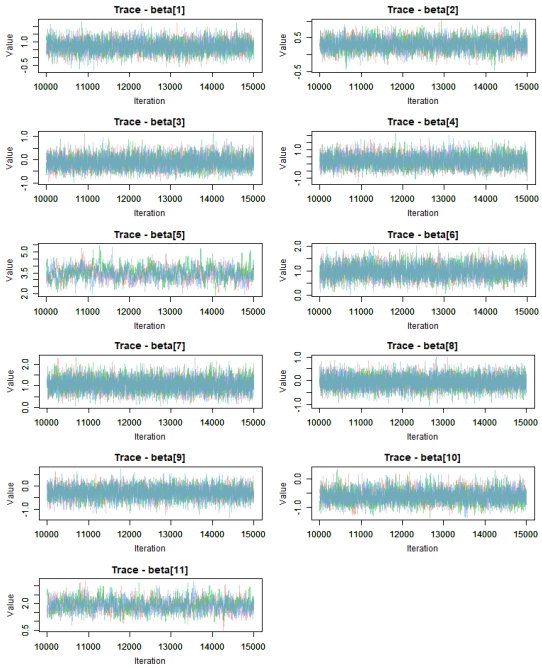

Fig. 5. Trace Plots of Beta Coefficients

To assess the trade-off between goodness-of-fit and model complexity, we calculated three standard Bayesian criteria:

- DIC (Deviance Information Criterion): The calculated DIC value was 229.7. This metric penalizes model complexity based on the effective number of parameters, serving as a classical Bayesian measure of fit.
- WAIC (Watanabe-Akaike Information Criterion): The WAIC was 247.2.
- LOO (Leave-One-Out Cross-Validation): The LOO metric was 248.0.

The convergence of these three metrics (DIC, WAIC, and LOO) to similar low values suggests that the model is robust, parsimonious, and does not suffer from influential outliers that often distort predictive accuracy in smaller medical datasets.

D. Comparison with Classical Methods

To validate the robustness of the Bayesian estimates, we compared the results with a standard Frequentist Logistic Regression (GLM) based on Maximum Likelihood Estimation (MLE). The comparison of Odds Ratios (OR) is presented in Table II.

TABLE II. COMPARISON OF ODDS RATIOS: BAYESIAN VS. FREQUENTIST

Variable	Bayesian OR	Frequentist (GLM) OR	Difference
HbA1c	30.528	25.625	4.903
BMI	6.799	5.947	0.852
TG	2.862	2.728	0.134
Chol	2.630	2.436	0.194
Gender	2.059	2.023	0.036
AGE	1.346	1.325	0.021
Cr	1.241	1.164	0.077
HDL	0.946	0.957	0.011
Urea	0.891	0.886	0.005

Variable	Bayesian OR	Frequentist (GLM) OR	Difference
LDL	0.759	0.777	0.018
VLDL	0.535	0.532	0.003

As shown in Table II, both frameworks exhibit a high degree of concordance regarding the *direction* of the effects. Variables identified as significant risk factors in the Bayesian model (HbA1c, BMI, TG, Chol) are also the strongest predictors in the Frequentist model. This confirms that the identified biomarkers are robust and not artifacts of a specific statistical methodology.

A notable numerical difference is observed in the strongest predictor, HbA1c (Bayesian OR: 32.0 vs. Frequentist OR: 25.6). While both values indicate an extreme risk of association, the divergence highlights a fundamental difference in estimation. The Frequentist MLE provides a point estimate that maximizes the likelihood function, whereas the Bayesian estimate averages over the entire posterior distribution. In regions of high non-linearity (extreme odds ratios), the asymmetry of the posterior distribution can lead to slightly higher mean estimates compared to the mode (MLE). However, clinically, both models lead to the identical conclusion: HbA1c is the dominant driver of diabetes status.

While the numerical estimates are comparable for most variables (e.g., VLDL differs by only 0.003), the preference for the Bayesian framework in this study—as argued in the Literature Review by Sullivan et al. [5]—is justified by interpretability. The Frequentist model relies on asymptotic normality assumptions to generate Confidence Intervals, which can be unreliable for skewed data. In contrast, the Bayesian Credible Intervals (presented in Section ii) are exact derivations from the data and priors, allowing us to intuitively state the probability of risk falling within a specific range. This feature is particularly valuable for "borderline" variables where clinical decision-making requires a nuanced understanding of uncertainty rather than a simple binary "significant/non-significant" label.

V. Conclusion & Future Work

A. Summary of Findings

This study successfully developed and validated a Bayesian Logistic Regression framework for the risk stratification of Type 2 Diabetes Mellitus. By leveraging a probabilistic approach, we addressed the critical need for interpretable medical AI models, moving beyond the "black-box" limitations of standard machine learning algorithms. The research yielded three primary conclusions:

1) Predictive Excellence: The Bayesian model achieved a classification accuracy of 96.30% and an AUC of 0.9754, outperforming recent ensemble learning benchmarks cited in the literature. Furthermore, the estimates showed strong concordance with classical Frequentist methods, confirming that the Bayesian framework maintains rigorous statistical validity while offering superior interpretability through probabilistic risk quantification.

2) Hierarchical Importance of Biomarkers: Consistent with global epidemiological data, HbA1c was identified as the paramount predictor of diabetes risk (OR: 30.88), followed by BMI and lipid markers (Triglycerides and Cholesterol). The Bayesian 95% Credible Intervals provided a precise quantification of the uncertainty surrounding these high-impact variables.

3) Clarification of Renal Markers: Contrary to emerging hypothesis in recent univariate studies, our multivariate analysis revealed that Urea and Creatinine are not significant independent predictors (OR ~ 1.0) when glycemic and anthropometric factors are controlled for. This highlights the model's ability to "explain away" redundant variance, suggesting that renal dysfunction in this cohort is likely a secondary comorbidity rather than a primary predictive signal for diabetes onset.

B. Limitations of the Study

While the findings are robust, this study is subject to specific limitations:

1) Geographic and Cohort Bias: The dataset represents a single-center cohort from Rashid Hospital (Baghdad, Iraq). Consequently, the posterior estimates (particularly the extreme Odds Ratio for HbA1c) may be specific to the genetic and environmental profile of this population and may not generalize perfectly to European or East Asian populations without re-calibration.

2) Prior Specification: To provide a fair comparison with classical Frequentist methods, this study utilized Non-Informative (Vague) Priors. While this validated the data-driven power of the model, it did not fully exploit the unique Bayesian capability of incorporating external expert knowledge (Informative Priors) to potentially stabilize estimates further.

3) Cross-Sectional Design: The data represents a snapshot of patient health. As such, the associations found are correlational, not causal. The model cannot definitively determine the temporal sequence between lipid dysregulation and hyperglycemia.

C. Recommendations for Future Research

To advance the clinical utility of this framework, future investigations should focus on:

1) Integration of Informative Priors: Future work should incorporate "Informative Priors" derived from large-scale meta-analyses (such as the IDF Diabetes Atlas). Comparing the performance of an Informative model against the Non-Informative baseline established in this study could demonstrate the true value of "cumulative learning" in medical statistics.

2) External Validation: The model should be tested on diverse external datasets (e.g., NHANES or PIMA Indians databases) to assess cross-population generalizability and robustness.

3) Bayesian Variable Selection: Implementing advanced techniques such as the Bayesian LASSO or Horseshoe Priors could provide a more automated and rigorous method for handling multicollinearity among lipid profiles (LDL, HDL, Cholesterol), potentially identifying a minimal optimal subset of biomarkers for cost-effective screening.

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Data Availability: Dataset is available from the Mendeley Data Repository, DOI: 10.17632/wj9rwkp9c2.1

CRediT authorship contribution statement

Filipus Darren Siswanto: Software, Methodology, Formal analysis, Writing – Original Draft, Visualization. **Sergio Winnero:** Conceptualization, Methodology, Data Curation, Investigation, Formal analysis, Writing – Original Draft, Project administration. **Hans Christian Soebroto:** Conceptualization, Software, Resources, Formal analysis, Writing – Review & Editing. **Nathanael Billy Christiano:** Investigation, Validation, Writing – Review & Editing. **Jaffarus Sodiq:** Methodology, Validation, Writing – Review & Editing, Supervision, Project administration.

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